

Index for assessing water quality in the Chone River, Manabí, Ecuador

Índice para avaliar a qualidade da água do Rio Chone, Manabí, Equador

Índice para evaluar la calidad del agua del Río Chone, Manabí, Ecuador

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Abstract

Water is an essential natural resource for the survival of living beings and the development of societies; however, increasing natural and anthropogenic pressures have contributed to the progressive deterioration of its quality. In this context, the objective of this study was to evaluate the water quality of the Chone River by applying the National Sanitation Foundation Water Quality Index (ICA-NSF), with the aim of determining its level of contamination and establishing its potential implications for the various uses of this water resource. To this end, five sampling points (P1–P5) located in the watershed associated with Chone, Bolívar, Tosagua, and Junín were analyzed, considering two sampling depths (30 and 60 cm). The calculation of the ICA-NSF incorporated nine variables: dissolved oxygen, fecal coliforms, pH, biochemical oxygen demand (BOD₅), temperature change, phosphates, nitrates, turbidity, and total solids, applying the index's classic weighting structure. The values obtained ranged from 31.4 to 66.2, with a predominance of the medium and poor quality categories. Point P1 exhibited the best relative conditions at both depths, while P3 recorded the highest degree of deterioration, primarily associated with high concentrations of nutrients, BOD₅, turbidity, and microbiological contamination. The comparison between depths revealed minor variations between 30 and 60 cm, suggesting a relatively homogeneous distribution of deterioration throughout the water column. In conclusion, the results indicate a water system under pressure from organic pollution and nutrient enrichment, with potential effects on ecological integrity and restrictions on the most sensitive uses of the resource, highlighting the need to strengthen strategies for monitoring, control, and management of water quality in the Chone River basin.

Keywords: ICA-NSF; Water quality; Nutrients; Turbidity; Microbiological contamination.

Resumo

A água constitui um recurso natural essencial para a sobrevivência dos seres vivos e para o desenvolvimento das sociedades; no entanto, as pressões crescentes de origem natural e antropogénica têm contribuído para a deterioração progressiva da sua qualidade. Neste contexto, o objetivo do presente estudo foi avaliar a qualidade da água do rio Chone através da aplicação do Índice de Qualidade da Água da National Sanitation Foundation (ICA-NSF), com o intuito de determinar o seu nível de poluição e estabelecer as suas possíveis implicações para as diferentes utilizações do recurso hídrico. Para tal, foram analisados cinco pontos de amostragem (P1–P5) localizados no sistema hídrico associado a Chone, Bolívar, Tosagua e Junín, considerando duas profundidades de amostragem (30 e 60 cm). O cálculo do ICA-NSF integrou nove variáveis: oxigénio dissolvido, coliformes fecais, pH, demanda bioquímica de oxigénio (DBO₅), variação de temperatura, fosfatos, nitratos, turbidez e sólidos totais, aplicando a estrutura clássica de ponderação do índice. Os valores obtidos oscilaram entre 31,4 e 66,2, com predominância das categorias de qualidade média e má. O ponto P1 apresentou as melhores condições relativas em ambas as profundidades, enquanto o P3 registou o maior grau de deterioração, associado principalmente a elevadas concentrações de nutrientes, DBO₅, turbidez e contaminação microbiológica. A comparação entre profundidades revelou variações reduzidas entre 30 e 60 cm, o que sugere uma distribuição relativamente homogénea da deterioração na coluna de água. Em conclusão, os

resultados revelam um sistema hídrico sujeito a pressão devido à poluição orgânica e ao enriquecimento de nutrientes, com efeitos potenciais sobre a integridade ecológica e restrições aos usos mais sensíveis do recurso, o que realça a necessidade de reforçar as estratégias de monitorização, controlo e gestão da qualidade da água na bacia hidrográfica do rio Chone.

Palavras-chave: ICA-NSF; Qualidade da Água; Nutrientes; Turbidez; Contaminação microbiológica.

Resumen

El agua constituye un recurso natural esencial para la supervivencia de los seres vivos y el desarrollo de las sociedades; sin embargo, las crecientes presiones de origen natural y antropogénico han contribuido al deterioro progresivo de su calidad. En este contexto, el objetivo del presente estudio fue evaluar la calidad del agua del río Chone mediante la aplicación del Índice de Calidad del Agua de la National Sanitation Foundation (ICA-NSF), con el propósito de determinar su nivel de contaminación y establecer sus posibles implicaciones para los diferentes usos del recurso hídrico. Para ello, se analizaron cinco puntos de muestreo (P1–P5) ubicados en el sistema hídrico asociado a Chone, Bolívar, Tosagua y Junín, considerando dos profundidades de muestreo (30 y 60 cm). El cálculo del ICA-NSF integró nueve variables: oxígeno disuelto, coliformes fecales, pH, demanda bioquímica de oxígeno (DBO₅), cambio de temperatura, fosfatos, nitratos, turbidez y sólidos totales, aplicando la estructura clásica de ponderación del índice. Los valores obtenidos oscilaron entre 31,4 y 66,2, con predominio de las categorías de calidad media y mala. El punto P1 presentó las mejores condiciones relativas en ambas profundidades, mientras que P3 registró el mayor grado de deterioro, asociado principalmente con elevadas concentraciones de nutrientes, DBO₅, turbidez y contaminación microbiológica. La comparación entre profundidades evidenció variaciones reducidas entre 30 y 60 cm, lo que sugiere una distribución relativamente homogénea del deterioro en la columna de agua. En conclusión, los resultados evidencian un sistema hídrico sometido a presión por contaminación orgánica y enriquecimiento de nutrientes, con potenciales efectos sobre la integridad ecológica y restricciones para los usos más sensibles del recurso, lo que resalta la necesidad de fortalecer las estrategias de monitoreo, control y gestión de la calidad del agua en la cuenca del río Chone.

Palabras clave: ICA-NSF; Calidad del agua; Nutrientes; Turbidez; Contaminación microbiológica.

1. Introduction

Water is a natural resource essential for the survival of living beings and the development of societies (Campos, 2019). This resource is not only vital for human consumption but also for economic, agricultural, and industrial activities (Pan American Health Organization [PAHO], 2024). However, globally, factors such as population growth, uncontrolled urbanization, industrial and agricultural pollution, and the effects of climate change have placed increasing pressure on water resources, significantly degrading their quality. This poses a serious threat to public health, given that approximately 1.7 billion people consume water contaminated with fecal matter (World Health Organization [WHO], 2023).

According to the World Water Assessment Programme, more than 80% of the wastewater generated worldwide is discharged into water bodies without prior treatment, thus posing a direct threat to aquatic ecosystems and public health (World Water Assessment Programme [WWAP], 2017). This situation has led to an alarming decline in freshwater quality, which is now considered one of the major global environmental challenges (United Nations Educational, Scientific and Cultural Organization [UNESCO], 2023). In Latin America, millions of people are exposed to health risks from consuming water contaminated with physical, chemical, and biological agents (PAHO, 2024).

In Ecuador, water resources face increasing pressure from domestic, agricultural, and industrial discharges that compromise their quality (Cárdenas Novillo, 2020). Durán et al. (2020) also highlight the impact of these activities on multiple watersheds across the country. According to the 2024 report on Environmental-Economic Information in Municipal GADs, 56.7% of untreated wastewater disposal sites discharged into rivers and 35.8% into streams (Instituto Nacional de Estadística y Censos [INEC], 2024). In the Chone River basin, located in Manabí, Solórzano & Quiroz (2021) documented that agriculture, livestock farming, and urban growth affect water quality.

Recent research has identified high concentrations of nutrients such as nitrogen and phosphorus in the waters of the Chone River. According to the study by Véliz & Seni (2022), total phosphorus levels range from 1.71 to 3.81 mg/L, while total nitrogen ranges from 5.88 to 14.00 mg/L, thus exceeding the limits established for unrestricted use, which indicates

eutrophication processes that compromise water quality and threaten aquatic biodiversity. This evidence highlights the urgency of implementing tools that allow for the effective assessment and monitoring of water quality.

Given the need for tools that allow for a comprehensive assessment of the condition of water bodies, the Water Quality Index (WQI) proposed by the National Sanitation Foundation (NSF) has been developed, as this index integrates physical, chemical, and microbiological parameters into a single numerical value that synthesizes the information obtained, thereby facilitating data interpretation and decision-making in water resource management (Paucar et al., 2023).

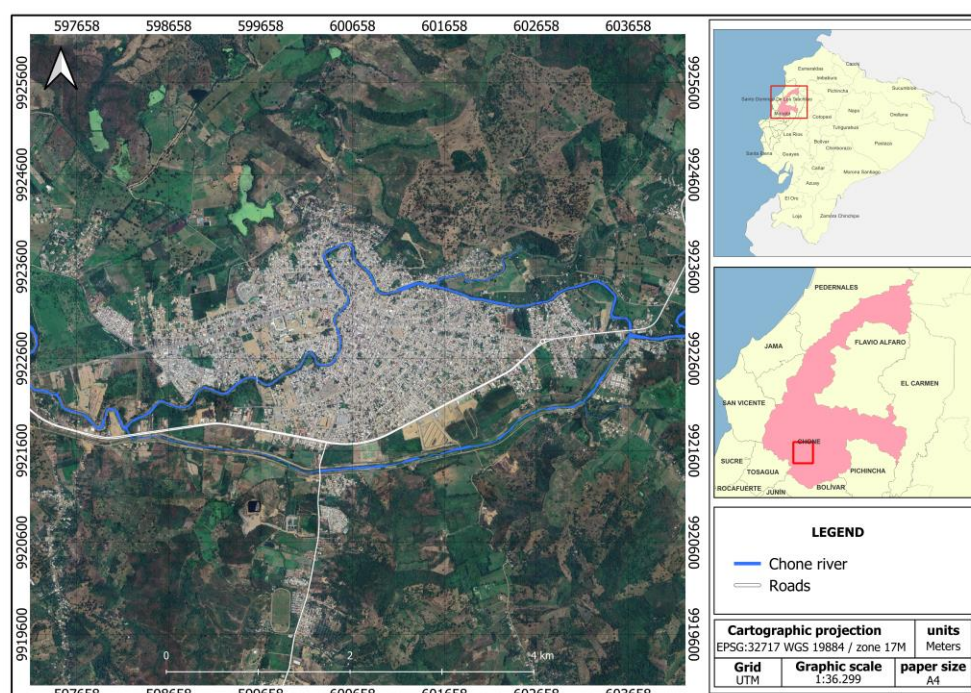
The Organic Code of the Environment (COA) promotes the sustainable use of watersheds and the development of technologies to protect biodiversity (Ministry of the Environment, Water, and Ecological Transition [MAATE], 2017). Likewise, the Sustainable Development Goals (SDGs), in Goal 6, seek to ensure clean water and sanitation, establishing targets to reduce water pollution and protect related ecosystems (United Nations [UN], 2018).

This study aims to assess the water quality of the Chone River by applying the Water Quality Index (WQI), to determine its level of pollution and its impact on the various uses of this water resource.

2. Methodology

An experimental field study was conducted (Risemberg et al., 2026a) using a quantitative approach (Risemberg et al., 2026b; Pereira et al., 2018) and simple descriptive statistics, including column charts, line charts, data classes, absolute frequency values (counts), and mean values (Shitsuka et al., 2014a; Shitsuka et al., 2014b). The study was conducted on the Chone River in the province of Manabí (Figure 1). Its watershed primarily encompasses the cantons of Chone, Bolívar, Tosagua, and Junín. This river originates in the western foothills of the Andes Mountains and flows southwest until it empties into the city of Bahía de Caráquez. This tributary plays a fundamental role in the agricultural, economic, and ecological development of the area, as it is one of the province's main waterways.

Figure 1. Map showing the geographic location of the study area.



Source: Google Earth (2025).

The study adopted a quantitative, non-experimental approach with an exploratory-descriptive scope, aimed at identifying water pollution resulting from wastewater discharges and agricultural waste, comparing the results obtained with the standards defined in the current regulatory framework.

Assessment of the Current State and Site Survey

The assessment of the current state of the Chone River began with an on-site survey aimed at identifying the main anthropogenic activities with a potential impact on water resource degradation. The study includes direct measurements of critical parameters such as turbidity, suspended solids (SS), total solids (TS), pH, phosphates, biochemical oxygen demand (BOD₅), temperature, fecal coliforms, nitrates, and dissolved oxygen (DO). These measurements are linked to the river's flow regime, a determining factor in the dynamics and health of aquatic ecosystems (Lara and Vélez, 2022). The monitoring stations were georeferenced using a portable GPS receiver, allowing for the precise mapping of strategic sites in spatial analysis tools such as Google Earth for subsequent territorial analysis (Orrala De La Cruz, 2024).

Water Quality Assessment

The ICA-NSF synthesizes multiple physicochemical and microbiological variables into a single value ranging from 0 to 100. Its original structure was proposed by Brown et al. (1970) and remains one of the most widely used tools for the integrated assessment of surface waters. The index was calculated using a weighted aggregation of quality sub-indices (Q_i) obtained from the method's standard rating curves, according to the general expression $ICA-NSF = \sum(Q_i \cdot W_i)$ (Coronel and Jaramillo, 2022; Quiroz et al., 2017).

Nine parameters were used in this study: dissolved oxygen (% saturation), fecal coliforms, pH, BOD₅, temperature change (ΔT), phosphates, nitrates, turbidity, and total dissolved solids (TDS). The standard weights of the method were adopted: DO = 0.17; fecal coliforms = 0.16; pH = 0.11; BOD₅ = 0.11; temperature = 0.10; phosphates = 0.10; nitrates = 0.10; turbidity = 0.08; total solids = 0.07 (Uddin et al., 2021; Gupta and Gupta, 2021; Kumar et al., 2024).

It should be noted that the database reports phosphates and TDS, whereas the original formulation of the index was based on total phosphorus and total solids. This substitution is common in applied practice, although it introduces a source of methodological uncertainty that must be explicitly stated when interpreting the results. The classification adopted was: 91–100 excellent, 71–90 good, 51–70 average, 26–50 poor, and 0–25 very poor.

Tables 1 and 2 summarize the experimental data collected at depths of 30 cm and 60 cm.

Table 1. Experimental database at 30 cm.

Parameter	P1	P2	P3	P4	P5
pH	8.21	8.09	8.10	8.09	7.92
ΔT (°C)	2.80	3.40	2.50	2.50	1.0
Nitrates (mg/L)	1.80	4.50	12.50	10.20	11.0
Phosphates (mg/L)	1.40	2.20	10.30	12.40	12.60
TDS (mg/L)	747.0	756.0	775.0	775.0	775.0
OD (% sat.)	109.5	141.8	120.5	108.4	120.5
BOD ₅ (mg/L)	3.30	65.0	23.0	5.0	31.0
Turbidity (NTU)	198.7	219.0	182.4	193.2	197.8
Fecal coliforms (MPN/100 mL)	21.0	43.0	240.0	93.0	43.0

Source: Research data (2026).

Table 2. Experimental database at 60 cm.

Parameter	P1	P2	P3	P4	P5
pH	8.08	8.03	8.04	8.02	8.04
ΔT (°C)	3.0	3.60	2.50	2.50	1.10
Nitrates (mg/L)	2.50	3.80	14.0	9.50	13.20
Phosphates (mg/L)	1.70	1.50	12.90	10.50	15.20
TDS (mg/L)	742.0	754.0	776.0	783.0	776.0
OD (% sat.)	142.9	129.6	120.5	120.5	132.4
BOD ₅ (mg/L)	11.10	57.0	22.0	5.10	34.80
Turbidity (NTU)	197.5	220.4	180.1	195.0	195.3
Fecal coliforms (MPN/100 mL)	9.0	21.0	43.0	43.0	21.0

Source: Research data (2026).

3. Results

The ICA values calculated for depths of 30 cm and 60 cm at each monitoring point are summarized in Table 3.

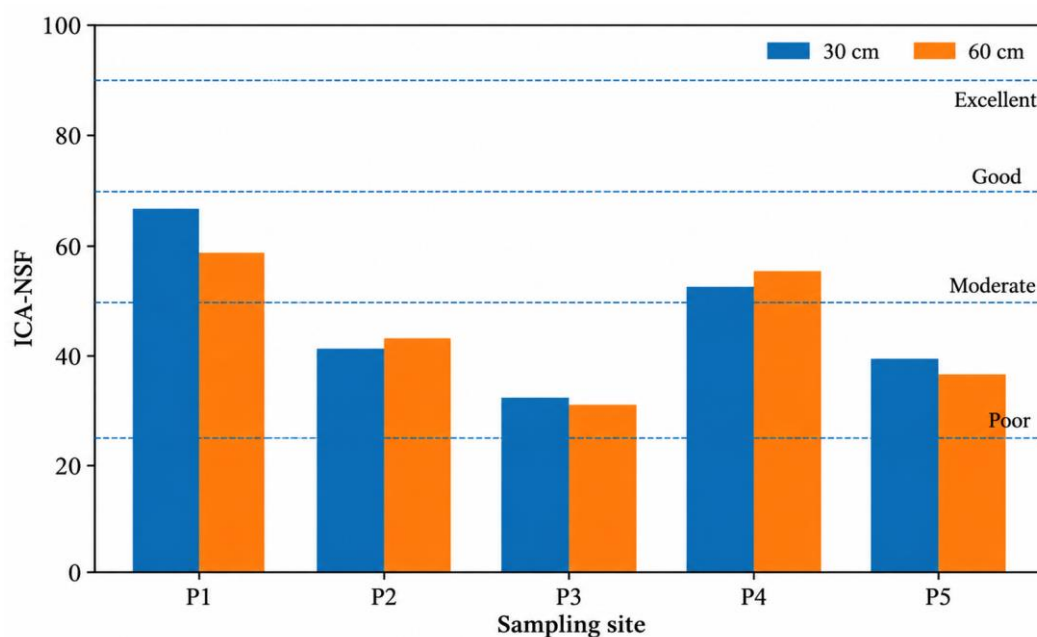
Table 3. ICA Results – NSF.

Point	ICA 30 cm	Class 30 cm	ICA 60 cm	Class 60 cm	ΔICA (60-30)	Average
P1	66.2	Median	58.7	Median	-7.5	62.5
P2	41.5	Poor	43.1	Poor	1.6	42.3
P3	32.8	Poor	31.4	Poor	-1.4	32.1
P4	52.6	Median	55.3	Median	2.7	54.0
P5	38.9	Poor	36.7	Poor	-2.2	37.8

Source: Research data (2026).

On average, the ICA-NSF was 46.40 at 30 cm and 45.04 at 60 cm, confirming an overall unfavorable quality. The best relative performance was observed at P1 (average = 62.45), while the worst performance was observed at P3 (average = 32.10).

Figure 2. Comparison of the ICA-NSF by sampling point at depths of 30 and 60 cm.



Source: Research data (2026).

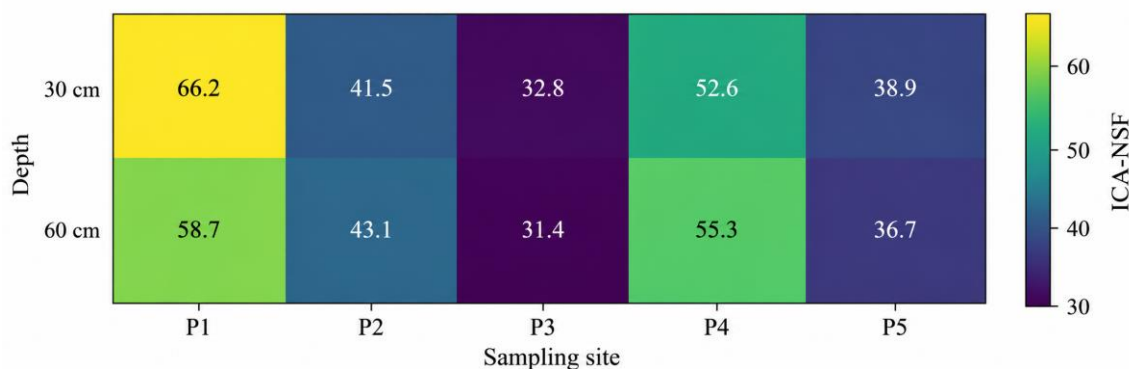
Figure 2 presents a comparison of the Water Quality Index at the five sampling points evaluated at depths of 30 and 60 cm. Overall, the results showed that water quality remained in the moderate to poor categories, with no values falling into the good or excellent classifications. The overall average obtained was 46.40 at a depth of 30 cm and 45.04 at 60 cm, reflecting an unfavorable environmental condition, which indicates the presence of physicochemical and microbiological alterations in the evaluated system.

Site P1 had the highest ICA-NSF values at both depths, with 66.2 at 30 cm and 58.7 at 60 cm, placing it in the “moderate” category. This pattern indicates that, although the site maintains relatively better conditions compared to the other sites, certain parameters limit the achievement of higher water quality. Factors contributing to this result included low concentrations of nitrates, phosphates, and fecal coliforms, as well as a moderate BOD₅ at 30 cm (3.30 mg/L). However, the decrease in the index at 60 cm is related to the increase in BOD₅ and temperature variation, which indicates a lower prevalence of optimal conditions at depth.

On the other hand, points P2, P3, and P5 fell into the “poor” category at both depths, indicating a significant deterioration in water quality. Point P3 recorded the lowest average value in the study (32.1), primarily associated with high concentrations of nitrates and phosphates, as well as high levels of fecal coliforms at 30 cm. These results indicate a significant influx of organic matter and nutrients, possibly related to wastewater discharges or diffuse pollution processes. Likewise, high turbidity and total dissolved solids negatively contributed to the index, reducing environmental quality conditions at this point.

In contrast, point P4 showed intermediate quality values, with indices of 38.9 and 36.7. This result was related to high nutrient concentrations—especially phosphates and nitrates—as well as considerable BOD₅ and turbidity values. Although fecal coliform concentrations were moderate, the combination of the other parameters significantly reduced the ICA-NSF value.

Figure 3. ICA-NSF heat map. The persistence of low values at P2, P3, and P5 reflects a consistent deterioration across depths.



Source: Research data (2026).

Figure 3 shows the distribution of the ICA-NSF using a heatmap, a tool that allows for the visual interpretation of the intensity and uniformity of water quality as a function of depth and sampling point. The heatmap facilitates the immediate identification of areas with similar characteristics through color variation, highlighting spatial patterns of environmental deterioration and stability.

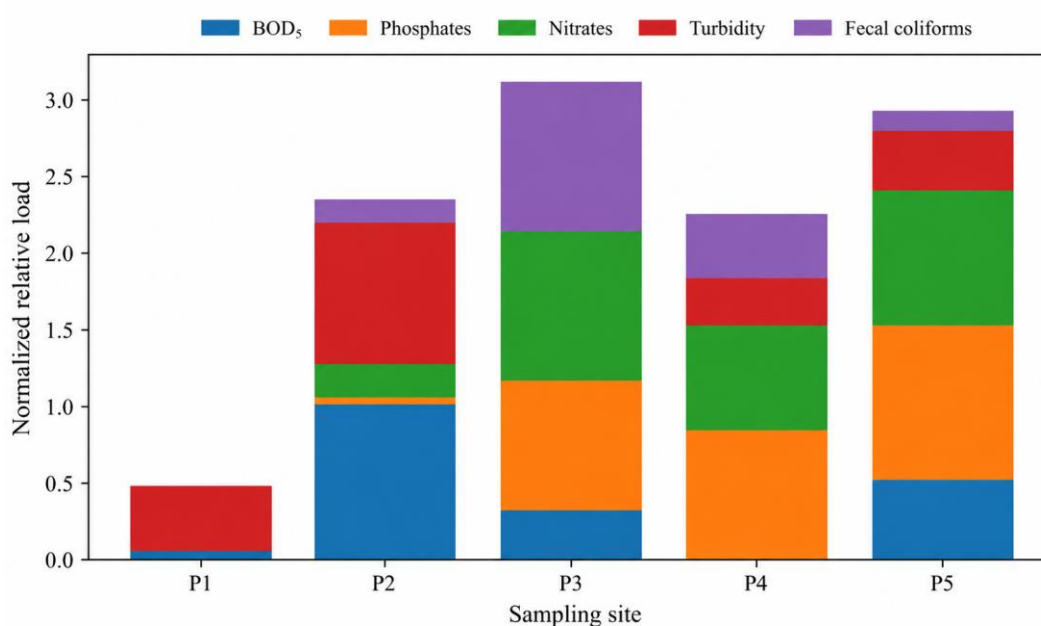
The color transition observed in the figure shows a marked contrast between the evaluated points. The yellow and green tones present at points P1 and P4 reflect relatively more favorable conditions, while the dark blue and violet tones at P2, P3, and P5 indicate areas with lower water quality. This color distribution demonstrates that environmental degradation was

not random but rather exhibits a distinct trend within the analyzed system.

One of the most relevant aspects shown in the figure is the visual similarity between the two evaluated depths. At most points, the colors remained consistent with minimal variations between 30 and 60 cm, reflecting vertical stability in the ICA-NSF's behavior. This indicates that pollution or environmental alteration processes influenced the water column relatively uniformly, with no drastic differences between surface and deeper levels.

Furthermore, the matrix clearly identified persistent critical areas. Point P3 exhibited darker shades at both depths, indicating consistently unfavorable conditions and visually standing out as the area most affected by environmental factors.

Figure 4. Normalized relative load of the main water quality stressors. The highest accumulation is concentrated at P3 and P5, explaining their lower performance in the integrated index.



Source: Research data (2026).

Figure 4 shows that the impact on water quality was not due to a single dominant pollutant, but rather to the combined interaction of various environmental stressors. However, the intensity and composition of these loads varied considerably among the sampling points, reflecting different types of pollution pressure within the analyzed system.

Point P1 had the lowest cumulative load of stressors, with mainly a moderate contribution from turbidity, while the remaining parameters showed low values compared to the other sites. This pattern indicates that the alterations at this point were primarily associated with the presence of suspended particles, with no strong evidence of organic or microbiological contamination.

On the other hand, point P2 showed a high load dominated primarily by BOD₅ and turbidity. The marked contribution of BOD₅ indicates a significant presence of biodegradable organic matter, suggesting possible inputs of organic waste or pollutant discharges. Likewise, the high contribution of turbidity reflects a considerable accumulation of suspended solids—factors that collectively contribute to a decline in water quality.

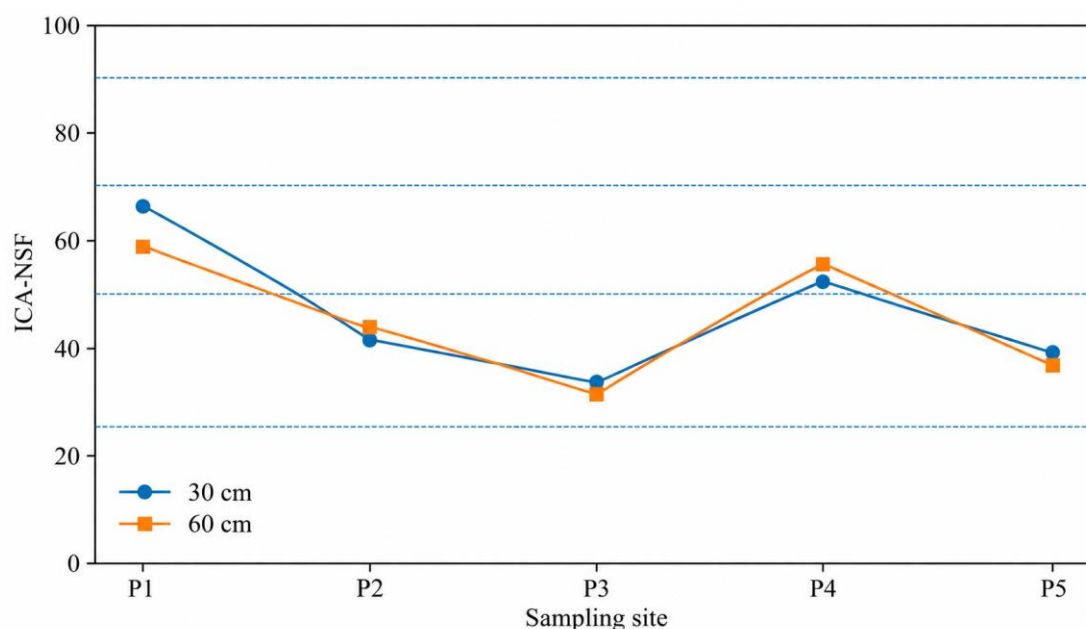
Point P3 exhibited one of the highest pollution loads in the study and stood out for its relatively balanced distribution of phosphate, nitrates, and fecal coliforms. This indicates a combination of nutrient-based pollutants and microbiological contamination, suggesting eutrophication processes and possible contributions from wastewater sources. Furthermore, the

significant presence of fecal coliforms makes this point one of the areas posing the greatest health risk within the evaluated region.

At sampling point P4, the relative load was primarily associated with phosphates and nitrates, while the contribution of BOD₅ was minimal. This pattern indicates that the deterioration at this site was more closely related to nutrient enrichment combined with a high organic load. The moderate presence of turbidity and fecal coliforms suggests moderate levels of contamination compared to the most critical sampling points.

Site P5 exhibited a high cumulative load, dominated by phosphates and nitrates, accompanied by a moderate contribution from BOD₅ and turbidity. The high proportion of nutrients indicates significant pressure associated with diffuse pollution processes or inputs of phosphorus- and nitrogen-rich compounds. Although fecal coliforms were relatively less prevalent, the combination of all stressors maintained conditions unfavorable for water quality.

Figure 5. Spatial profile of the ICA-NSF along the sampling points. The similarity between the curves suggests little vertical stratification of the deterioration.



Source: Research data (2026).

Figure 5 presents the spatial profile of the ICA-NSF along the sampling points evaluated at depths of 30 and 60 cm, allowing for the observation of longitudinal variation in water quality within the analyzed system. The behavior of both curves shows a similar trend across depths, indicating that environmental conditions maintained a relatively uniform spatial pattern throughout the study area.

In both profiles, a progressive decrease in the ICA-NSF was identified from P1 to P3, followed by a partial recovery at point P4 and a further reduction at P5. This pattern reflects the existence of sectors with different levels of environmental stress, where certain points acted as areas of greater deterioration and others exhibited relatively more favorable conditions.

Next, the ICA values obtained in this study are compared with the limits established in Ministerial Agreement 097-A, in order to determine the water body's degree of legal compliance.

Table 4. Comparison of the ICA values obtained with the limits set by current environmental regulations.

Parameter	P1		P2		P3		P4		P5		LMP	
	30 cm	60 cm	30 cm	60 cm	30 cm	60 cm	30 cm	60 cm	30 cm	60 cm	Freshwater	Saltwater
pH	8.21	8.08	8.09	8.03	8.10	8.04	8.09	8.02	7.92	8.04	6.5–9	6.5–9.5
ΔT (°C)	2.80	3.0	3.40	3.60	2.50	2.50	2.50	2.50	1.0	1.10	-	-
Nitrates (mg/L)	1.80	2.50	4.50	3.80	12.50	14.0	10.20	9.50	11.0	13.20	13	200
Phosphates (mg/L)	1.40	1.70	2.20	1.50	10.30	12.90	12.40	10.50	12.60	15.20	-	-
TDS (mg/L)	747.0	742.0	756.0	754.0	775.0	776.0	775.0	783.0	775.0	776.0	-	-
OD (% sat.)	109.5	142.9	141.8	129.6	120.5	120.5	108.4	120.5	120.5	132.4	> 80	> 60
BOD ₅ (mg/L)	3.30	11.10	65.0	57.0	23.0	22.0	5.0	5.10	31.0	34.80	20	-
Turbidity (NTU)	198.7	197.5	219.0	220.4	182.4	180.1	193.2	195.0	197.8	195.3	-	-
Fecal coliforms (MPN/100 mL)	21.0	9.0	43.0	21.0	240.0	43.0	93.0	43.0	43.0	21.0	-	-

Source: Research data (2026).

According to Table 4 of Ministerial Agreement 097-A, regarding acceptable quality criteria for the preservation of aquatic and wildlife in freshwater and marine environments, the results obtained at the various monitoring sites along the Chone River showed variable trends among the analyzed parameters.

pH values ranging from 7.92 to 8.21 remained within the permissible limits established by regulations for freshwater and saltwater, indicating favorable chemical conditions for the maintenance of aquatic ecosystems and the development of hydrobiological organisms. Likewise, dissolved oxygen showed saturation percentages ranging from 108.4% to 142.9%, far exceeding the minimum values required by regulations—which set >80% for freshwater and >60% for saltwater—reflecting adequate oxygenation capacity in the water body and favorable conditions for the respiration and survival of aquatic biota.

Regarding nitrate concentration, the values obtained ranged from 1.80 to 14.0 mg/L. According to regulations, the permissible limit for freshwater is 13 mg/L, indicating a slight noncompliance at point P3 at a depth of 60 cm, where a concentration of 14.0 mg/L was recorded. However, all values obtained remained below the limit established for saltwater. This isolated increase is associated with the input of nutrients from agricultural runoff, domestic discharges, and/or the decomposition of organic matter.

On the other hand, BOD₅ concentrations ranged from 3.30 to 65.0 mg/L. According to Table 3 of Ministerial Agreement 097-A, the permissible limit for freshwater is 20 mg/L; however, points P2, P3, and P5 exceeded this value, with P2 at depths of 30 cm and 60 cm standing out in particular, with concentrations of 65.0 mg/L and 57.0 mg/L, respectively. These results indicate a high load of biodegradable organic matter in certain sections of the river, a situation that may cause disturbances in the ecological balance, an increase in decomposition processes, and potential impacts on aquatic fauna.

Regarding the parameters of phosphates, total dissolved solids (TDS), turbidity, and fecal coliforms, Table 3 of Ministerial Agreement 097-A does not establish specific permissible limits for the preservation of aquatic life; however, the high concentrations of phosphates and turbidity observed at several points suggest the presence of sediment, nutrient, and organic matter inputs into the water body—factors that may contribute to eutrophication and the progressive deterioration of the river’s environmental quality.

4. Discussion

The results show a pattern of moderate to severe water quality deterioration. It is worth noting that no sampling site achieved the “good” or “excellent” categories; on the contrary, three of the five sites remained in “poor” condition at least at

one of the depths. This distribution suggests sustained anthropogenic disturbance rather than an isolated, one-time event. In quantitative terms, the recorded ICA–NSF values were 46.40 (30 cm) and 45.04 (60 cm), confirming an overall unfavorable water quality. These findings differ from those reported by Vera and Zambrano (2025) for the Canuto River, where “good” water quality was observed, with ranges between 77.5 and 81.9. However, the data align with the findings of Das (2026), who notes that the recurrence of “unsuitable” or “very poor” levels typically stems from anthropogenic activities such as the discharge of industrial effluents, agricultural runoff, and domestic wastewater.

The persistently low values at P2, P3, and P5 are consistent with water bodies subject to organic inputs, fine sediment transport, and nutrient enrichment—a combination typically associated with domestic discharges and runoff from developed areas. The most compromised site was P3, with an average of 32.10, which consistently places it in the “poor” category. This pattern is consistent with its combination of elevated nitrate levels (12.5–14.0 mg/L), very high phosphate levels (10.3–12.9 mg/L), BOD₅ exceeding 22 mg/L, and the highest recorded level of fecal coliforms at the surface (240 NMP/100 mL). From an ecological perspective, this convergence of pressures implies an environment prone to trophic imbalances, increased biological oxygen demand, and a loss of suitability for recreational use or water supply without robust treatment.

Regarding nutrient dynamics, Yap et al. (2020) argue that nitrate is essential for the growth of aquatic plants and phytoplankton; however, Halder et al. (2020) warn that high concentrations can lead to plant mortality and cause respiratory problems in humans. On the other hand, Al-haen et al. (2026) and Shah et al. (2025) note that, although phosphates are less mobile than nitrate, their transport into surface water bodies is a key factor in eutrophication processes. Similarly, the presence of phosphates may result from the application of organic fertilizers, whose concentration typically decreases as soil depth increases. The authors state that in their study, well depths ranged from 5 to 15 m. Missaoui et al. (2026) assert that the elevated levels recorded are attributable to the intensive and prolonged agricultural use of phosphate fertilizers.

P1 showed the best relative condition at both depths, although it remained at only an average quality level. This indicates that the system retains some areas with comparatively less impact, but they are not free from pressure. The simultaneous presence of turbidity near 200 NTU and TDS levels exceeding 740 mg/L confirms that even the most favorable site still operates under a significant load of suspended and dissolved solids, which limits the interpretation of a genuine recovery of the water body. Atuhaire et al. (2026) argue that higher turbidity levels can be attributed to high biological activity, including the decomposition of organic matter and runoff from agricultural fields into water bodies. For their part, authors such as Chaves (2025) and Silva et al. (2015) state that suspended or dissolved matter in water—typically composed of minerals, metals, and salts—has a detrimental effect on the quality of water bodies because it causes turbidity and promotes heat absorption, thereby increasing the temperature.

The variation in depth was minimal. The overall average decreased only slightly from 46.40 to 45.04, and the specific variation ranged from -7.5 to 2.7 units. This behavior suggests sufficient vertical mixing to distribute the pollutant load throughout the water column. In terms of management, this means that the deterioration should not be interpreted as a disturbance restricted to the surface layer, but rather as a condition affecting the system as a whole.

From a methodological standpoint, the ICA–NSF is particularly useful due to its ability to integrate heterogeneous variables into a single, comprehensible scale; however, recent literature cautions that its performance depends on the consistency between the original parameters and the surrogates used in real-world datasets. Therefore, in this study, the interpretation focuses on the comparative and ecological value of the index, rather than on an absolute normative reading. Even with this caveat, the signal of deterioration is robust, as multiple critical parameters point in the same direction. Gil et al. (2018) indicate that an ICA study can serve as a guide for local authorities in making appropriate decisions to implement corrective measures.

Overall, the evidence points to a body of water subject to organic and eutrophication-causing pollution, exacerbated by high turbidity and microbiological contamination. To strengthen the discussion in a version submitted for peer review, it would be advisable to supplement these results with information on land use, the hydrological regime, the identification of discharge sources, and comparisons with national or regional standards applicable to the type of resource use.

5. Conclusion

It is concluded that the water body is in a state of critical degradation, with an ICA–NSF ranging from 31.4 to 66.2, falling predominantly into the “moderate” to “poor” quality categories. The deterioration is particularly severe at point P3, where the high concentration of nutrients and organic matter indicates a persistent source of contamination. The consistency of these results at depths of 30 and 60 cm confirms that the disturbance affects the entire water column, which validates the hypothesis of sustained anthropogenic pressure. Given this scenario, it is imperative to implement a comprehensive management plan that includes rigorous control of discharges and runoff management systems to halt the progression of water quality deterioration.

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